

Case-Volume Competency System

Replace Calendar-Based APP Onboarding with Evidence-Driven Surgical Development

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The MBA PA

A framework for cardiac surgery programs to define readiness by case exposure — not calendar duration. Includes case complexity classification, weighted volume targets, critical case-type checklists, and structured competency gates that produce autonomous APPs faster and more consistently.

Why Calendar-Based Onboarding Produces Uneven Results — Every Single Time

Time-based onboarding doesn't fail because the timeline is wrong. It fails because time is the wrong unit of measurement for surgical competence.

WHAT THE CALENDAR SAYS	WHAT THE CASES ACTUALLY DELIVERED
"90 days of cardiac surgery onboarding complete"	APP shadowed 32 CABGs, 4 valves. Never managed a bleeding patient. Never cannulated independently. Never saw femoral-femoral bypass.
"Six-month competency evaluation passed"	Evaluation marked "satisfactory" by a surgeon who worked with this APP on 4 cases total. Sample too small to mean anything.
"One-year mark — independent OR assignment"	APP independent on veins for 6 months. Never harvested a radial. Never managed conduit spasm. First independent assignment pairs them with a surgeon who doesn't know their gaps.
"Standardized 12-month program for all APPs"	Three APPs hired 2 months apart: one plateaued at month 4 from boredom (came from high-volume center), one needed 2x exposure (came from low-acuity hospital), one ready at month 7. All treated identically.

The calendar doesn't measure experience. It measures attendance. And in cardiac surgery, attendance is not a clinical credential.

1) Case Complexity Classification

Not all cases are equal. Volume targets are meaningless without complexity weighting.

TIER	DEFINITION	EXAMPLES	WEIGHT
Tier 1 — Routine	Standard procedure, stable patient, predictable course	Elective CABG x3-4, single valve AVR/MVR, stable hemodynamics, no prior sternotomy	1.0x
Tier 2 — Complex	Multiple procedures, moderate risk factors, challenging anatomy	CABG + valve, redo sternotomy, low EF (<35%), porcelain aorta, carotid + CABG	1.5x
Tier 3 — High-Acuity	Emergent/urgent, hemodynamically unstable, unusual procedure	Emergent CABG for STEMI, acute Type A dissection, post-arrest surgery, ECMO cannulation, cardiac trauma	2.0x

2) Weighted Case Volume Targets by APP Role

Define readiness in case-volume terms for each development stage.

DEVELOPMENT STAGE	WEIGHTED CASES	TIER 1 MIN	TIER 2 MIN	TIER 3 MIN	SURGEON DIVERSITY
Supervised Assist	40	25	10	3	Min 2 attendings
Transitional Assist	80	50	18	6	Min 3 attendings
Independent Assist	130	80	30	10	Min 4 attendings
Full Autonomy	180	110	45	15	Min 4 + emergent call

Key principle: A weighted case target means an APP who does mostly Tier 3 cases advances faster — appropriately — than one who does only Tier 1 electives. The weighting system rewards exposure to complexity, which builds the judgment that calendar-based onboarding assumes will appear on its own.

CASE-VOLUME COMPETENCY SYSTEM — PAGE 2

Critical Case-Type Checklists & The Living Case Log

3) Critical Case-Type Checklists

Certain experiences must appear in an APP's log before advancement. These separate someone who can assist from someone who can anticipate.

Before Independent Assist

- ☐ Cannulation: arterial — ≥ 10 cases
- ☐ Cannulation: venous (single-stage) — ≥ 10 cases
- ☐ Cannulation: femoral/axillary alternative access — ≥ 3 cases
- ☐ Cannulation: bicaval — ≥ 5 cases
- ☐ Chest closure — all layers — ≥ 15 cases
- ☐ Femoral-femoral bypass setup — ≥ 2 cases
- ☐ Coming off bypass — observation and assisted weaning — ≥ 8 cases
- ☐ Post-protamine hemostasis and chest exploration — ≥ 10 cases
- ☐ Redo sternotomy (assisted) — ≥ 1 case
- ☐ Intraoperative complication management — ≥ 1 case with documented attending debrief
- ☐ Chest re-opening for post-op bleeding — ≥ 3 cases
- ☐ Acute Type A dissection (assisted, incl. cooling and circulatory arrest) — ≥ 1 case

Before Full Autonomy (add to above)

- ☐ All above at independent level
- ☐ Emergent off-pump → on-pump conversion — ≥ 1 case
- ☐ ECMO cannulation or decannulation — ≥ 2 cases
- ☐ Cardiac trauma or penetrating injury (or simulated equivalent) — ≥ 1 case
- ☐ Independent intraoperative crisis management with attending present — ≥ 2 cases, documented

4) The Living Case Log Dashboard

The case log isn't a compliance document. It's the operating system of your APP development program.

What a Functional Case Log Captures (Per Case)

FIELD	WHY IT MATTERS	EXAMPLE
Date + OR room	Tracks density and spacing of exposure	2026-07-06, OR 4
Procedure(s)	Primary and secondary procedures	CABG x4 + LIMA→LAD, SVG→OM1, PDA, RCA
Case tier	Weighted volume tracking	Tier 2 (low EF 30%)
Attending surgeon	Diversity requirement tracking	Dr. Chen
APP role	Granular role description	Assisted cannulation + independent chest closure
Skills performed	Specific, not "assisted"	Arterial cannulation (Seldinger), proximal anastomosis assist, sternal wire closure
Notable events	Complications, learning moments	SVG kinking recognized pre-closure — revised with surgeon
Surgeon feedback	Captured same-day	"Good cannulation — work on earlier recognition of conduit length"
Self-assessment	APP's own reflection	Confident on cannulation. Need more reps on proximal anastomosis positioning.

Structured Competency Gates & APP Pace Management

5) Competency Gates — Advance When Evidence Says So

Every gate has four components: Volume threshold, Surgeon attestations, Performance evidence, Gate committee decision.

Gate I: Supervised → Transitional Assist

Volume threshold: 40 weighted cases (min 25 Tier 1, 10 Tier 2, 3 Tier 3) + critical checklist complete

Surgeon attestations (min 2 attendings):

- ☐ APP demonstrates safe sterile technique and OR protocol without prompting
- ☐ APP can verbally anticipate next 3-5 steps in a routine CABG sequence
- ☐ APP identifies and communicates abnormal findings without waiting to be asked
- ☐ APP cannulation technique is consistent and safe under direct supervision

Performance evidence: 5 consecutive cannulations with ≤ 1 correction required • 3 consecutive chest closures with no sterile field breach • Case log review confirms diversity requirement

Gate II: Transitional → Independent Assist

Volume threshold: 80 weighted cases (min 50 Tier 1, 18 Tier 2, 6 Tier 3) + critical checklist complete

Surgeon attestations (min 3 attendings, incl. ≥ 1 from Tier 2/3 cases):

- ☐ APP independently manages cannulation and decannulation without prompting on routine cases
- ☐ APP recognizes and communicates hemodynamic changes during bypass without waiting for direction
- ☐ APP independently manages chest closure including wire placement, hemostasis, and drain positioning
- ☐ APP demonstrates effective communication under pressure — no freezing, no information withholding

Performance evidence: 10 consecutive cannulations with zero corrections • ≥ 3 cases where APP independently recognized and escalated a surgical concern before attending • ≥ 1 complication managed with appropriate escalation and calibration

Gate III: Independent → Full Autonomy

Volume threshold: 130 weighted cases (min 80 Tier 1, 30 Tier 2, 10 Tier 3) + all critical checklists complete

Surgeon attestations (min 4 attendings, incl. ≥ 2 from different surgical groups):

- ☐ APP manages full operative sequence independently — anticipating surgeon needs, managing bypass transitions, directing the room during critical moments
- ☐ APP independently troubleshoots: cannulation difficulty, conduit issues, unexpected findings, bleeding — escalates appropriately
- ☐ APP functions as surgical team leader — directing perfusion, anesthesia communication, nursing workflow
- ☐ APP has demonstrated safe independent judgment in ≥ 2 emergent/urgent cases

Performance evidence: 30 consecutive independent cases with zero safety events • 2 documented crisis management events with attending confirmation • All critical case types independently performed • No unresolved trust flags or trust reset protocols active

Managing the Fast, Slow, & Stalled APP & Program Self-Assessment

6) APP Pace Management — One Size Does Not Fit All

The Fast APP — Accelerated Advancement

An APP arrives with 3 years of cardiac surgery experience from another center.

- ☐ **Day 1 case audit:** Map their prior log against your tier definitions and critical checklists. Where are the gaps?
- ☐ **Accelerated track:** Compress supervised phase to what's genuinely new — not a redo of what they already know.
- ☐ **Gap-focused assignment:** Target missing case types. If they've done 100 CABGs but never a femoral cannulation, that's priority one.
- ☐ **Gate eligibility:** Advance when they meet your program's criteria — not when the calendar says they've "done their time."

The Slow APP — Structured Support Without Limbo

An APP at 60 weighted cases — surgeon attestations aren't coming.

- ☐ **Gap diagnosis:** Technical? Judgment? Communication? The case log + surgeon feedback tells you.
- ☐ **Targeted remediation:** Not "observe 10 more cases" — specific goals. "Next 5 cannulations: zero guidewire errors, zero purse-string corrections."
- ☐ **Remediation pairing:** Assign to the surgeon who gives the most structured feedback — not the one who silently does the case and says it "didn't feel right."
- ☐ **Re-evaluation trigger:** "After 15 additional weighted cases with documented performance, re-submit for attestation." Not "we'll check in 3 months."

The Stalled APP — When Progress Plateaus

Same competency area stalled 4+ weeks. This is usually a scheduling failure, not a performance failure.

- ☐ **Dashboard alert:** Living case log flags any competency area with zero new cases in 30 days.
- ☐ **Scheduling intervention:** APP lead works with OR scheduling to prioritize missing case types and surgeon pairing.
- ☐ **Escalation if unresolved:** Gap persisting beyond 60 days despite scheduling intervention → program-level discussion.

7) Self-Assessment — Calendar or Cases?

Seven Dimensions of Competency-Driven Development

1 APP onboarding defined by case-volume targets, not calendar milestones

0 pts — Calendar-based. 1 pt — Some case targets. 2 pts — Weighted case-volume targets with tier classification drive all advancement. Score: ____ / 2

2 Critical case types identified and tracked for each APP

0 pts — No case-type tracking. 1 pt — Gaps known but no checklist. 2 pts — Written critical case-type checklist with stage-specific requirements. Score: ____ / 2

3 Case logs capture detailed skills, events, and feedback

0 pts — No logs or bare-minimum codes. 1 pt — Inconsistent logs. 2 pts — Living case logs with skills, events, surgeon feedback, self-assessment. Score: ____ / 2

4 Living dashboard shows each APP's progress against competency targets

0 pts — No dashboard. 1 pt — Dashboard exists but irregularly updated. 2 pts — Updated per case, reviewed weekly by APP lead, drives scheduling. Score: ____ / 2

5 Advancement uses structured gates

0 pts — "Surgeon says ready." 1 pt — Some documentation. 2 pts — Full gate packet: volume threshold, critical checklist, attestations, performance evidence, committee decision. Score: ____ / 2

6 Fast-track and remediation pathways exist

0 pts — Everyone follows same timeline. 1 pt — Informal accommodations. 2 pts — Written protocols for accelerated advancement, targeted remediation, stall intervention. Score: ____ / 2

7 Scheduling actively targets APP competency gaps

0 pts — APPs assigned by availability only. 1 pt — APP lead occasionally advocates. 2 pts — Systematic gap-driven scheduling — APP lead reviews dashboard weekly and coordinates with OR scheduling. Score: ____ / 2

Composite Score: ____ / 14

10–14 **Competency-Driven.** APPs developed by what they've done, not how long they've been employed. Surgeons trust the system because evidence is visible. APPs trust it because the path is clear.

5–9 **Transitioning.** Tracking cases but calendar is still primary decision tool. Advancement relies more on gut feel than structured evidence. Pieces are there — need to be connected into a system.

0–4 **Calendar-Dependent.** Time served, not experience earned. Fast APPs bored. Slow APPs invisible until evaluation day. Nobody knows who's actually ready for what.

What's Inside the Full Case-Volume Competency System Toolkit

- **Case complexity classification guide** — Tier 1/2/3 definitions for 25+ cardiac surgery procedure types with weighting rationale
- **Role-specific case-volume target calculator** — adjustable targets for first-assist, vein harvest, ICU-transition, and combined roles
- **Critical case-type checklist builder** — stage-specific templates with anatomical rationale
- **Living case log template** — structured format with skills, feedback, self-assessment, and dashboard auto-population
- **Competency gate packet templates** — Stage I→II→III→IV with volume thresholds, attestation forms, performance evidence requirements
- **Accelerated advancement protocol** — fast-track pathway with prior-case audit and gap-focused scheduling guide
- **Remediation and stall intervention protocol** — targeted remediation plan, re-evaluation triggers, scheduling escalation workflow

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This is an operational systems framework, not clinical teaching or patient-care directives. It draws on established principles of competency-based medical education applied to cardiac surgery workforce development.

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